

Visualization

Prof. Bernhard Schmitzer, Uni Göttingen, summer term 2025

Exam 2025-07-21

General remarks.

- The exam is built on three datasets provided in `data_X/`. Each dataset is briefly described in a corresponding meta file and the reference to the original source is given. Acquaint yourselves with all datasets and their conventions.
- Work with Python and jupyter notebooks, e.g. via the GWDG jupyter server at <https://jupyter-cloud.gwdg.de/>.
- Your submission should consist of (multiple) notebook files (optionally in text format, e.g. via jupyter text), combined into a single zip/tar archive.
- Do not include the data files in your submission.
- Submit via StudIP, similar to the problem sheets.
- **Submission deadline: 2025-08-01 18:00**
- Indicate your enrolment numbers clearly at the beginning of each notebook file.
- Groups of two students do not have to complete problem 1.
- To judge the expected amount of work in the assignments, an approximate number of suggested charts is given in parentheses. There is no single right solutions to the assignments. Sometimes more or even less charts may be appropriate, depending on your approach. **A single chart may consist of small multiples.** Including excessively many charts in your submission without a nuanced discussion will be considered as poor judgment.
- If you do not understand a problem description, do not hesitate to contact schmitzer@cs.uni-goettingen.de.

Problem 1: USA CDC suicide statistics.

This exercise deals with data on suicides in the US. We understand that struggles with mental health are prevalent throughout society and also among students. If you struggle with this and this exercise gets to you, please do not hesitate to seek professional help. A starting point might be the Psychosocial Counseling unit of the Studierendenwerk¹.

In this problem we work with statistics on suicide rates and the influence of gender, race, and age. The dataset is provided by the United States of America Centers for Disease Control and Prevention.

1. Import the csv data file `suicide.csv`. Filter out rows that contain missing values or values that are deemed statistically unreliable (see meta data). Filter out data from years earlier than 1980.

¹<https://www.studierendenwerk-goettingen.de/en/social-and-psychosocial-counselling/psb>

2. Show the evolution of suicide rates for each gender over the years by considering the rows of the dataset where `STUB_NAME=="Sex"`. Since this involves data aggregated over all age brackets (and the rate depends substantially on age), it is customary to compute an age-adjusted rate, which eliminates the effect of changing age distributions. Therefore, show both the crude and the age-adjusted data. (1 chart)
3. Now split the dataset according to gender and race/ethnicity by considering the rows of the dataset where `STUB_NAME=="Sex and race and Hispanic origin"` and show the evolution of rates over the years. Show again crude and age-adjusted data. Note that this (and all following sub-problems) involves careful parsing of the values in the column `STUB_LABEL`. (1 chart)
4. Now split the dataset according to gender and age by considering the rows of the dataset where `STUB_NAME=="Sex and age"` and show the evolution of rates over the years. Since here rates per age bracket are shown, no age-adjustment is required. (1-2 charts)
5. Now split the dataset according to gender, race, and age by considering the rows of the dataset where `STUB_NAME=="Sex, age and race and Hispanic origin"` and show the evolution of rates over the years. (1-2 charts)

Problem 2: Tourism accommodation statistics of Kärnten.

In this problem we work with statistics on arrivals and spent nights of tourists in the Austrian state of Kärnten (Carinthia).

1. Import the main csv data file `tourism.csv`.
2. Visualize the total number of nights spent by tourists from different origins. The auxiliary file `origin_category.csv` may be useful here. (1-2 charts)
3. Show the evolution of total spent nights over the years. How does the trend differ between domestic, German, European, and international tourists? (1-2 charts)
4. Show how many nights are spent in accommodations of different types. How does the choice of accommodation depend on origin? (2 charts)
5. The dataset is highly aggregated and only provides total arrivals (`Ankünfte`) and total spent nights (`Übernachtungen`) in each category (year, accommodation, origin). Still, taking the row-wise quotient of `Übernachtungen` by `Ankünfte` gives a rough estimate of the approximate duration of stays. Show how this duration depends on the type of accommodation. (1 chart)

Problem 3: Dynamics of biomolecule alanine dipeptide.

Alanine dipeptide is a very simple biomolecule that serves as a model system for analysing the dynamics of proteins and other biomolecules. In this problem we study computer simulated dynamics data of this molecule.

1. Import the three main arrays (positions, pairwise distances, angles) and the auxiliary data on atom types and atom bonds from `molecule.npz` into python.

2. Plot a simple 3D stick figure model of the atom positions of the molecule at $t = 0$ with an indication of the bonds. Then create a small animation that shows the evolution over the first 100 time steps. One should see slight ‘wiggling’ in most time steps, and occasional faster jumps. (2 charts)
3. Show the distribution of the dihedral angles (ϕ, ψ) throughout the whole trajectory. (1 chart)
4. Add an animation of the movement of (ϕ, ψ) over the first 100 time steps to the previous plot. (1 chart)
5. Apply PCA to the collection of pairwise distance vectors. Show the spectrum of eigenvalues. (1-2 charts)
6. Compute the projection coordinates of the (centered) distance vectors to the first few PCA eigenvectors. Show some selected 2-dimensional or 3-dimensional PCA embeddings. Is the PCA embedding consistent with the low-dimensional description by the dihedral angles? (2-3 charts)
7. From the atom position data, compute the lengths of all atom bonds at all times and show the distribution of each bond length over time. Which bond lengths are similar? (1 chart)